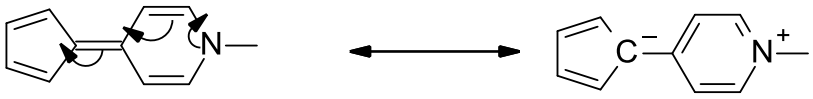
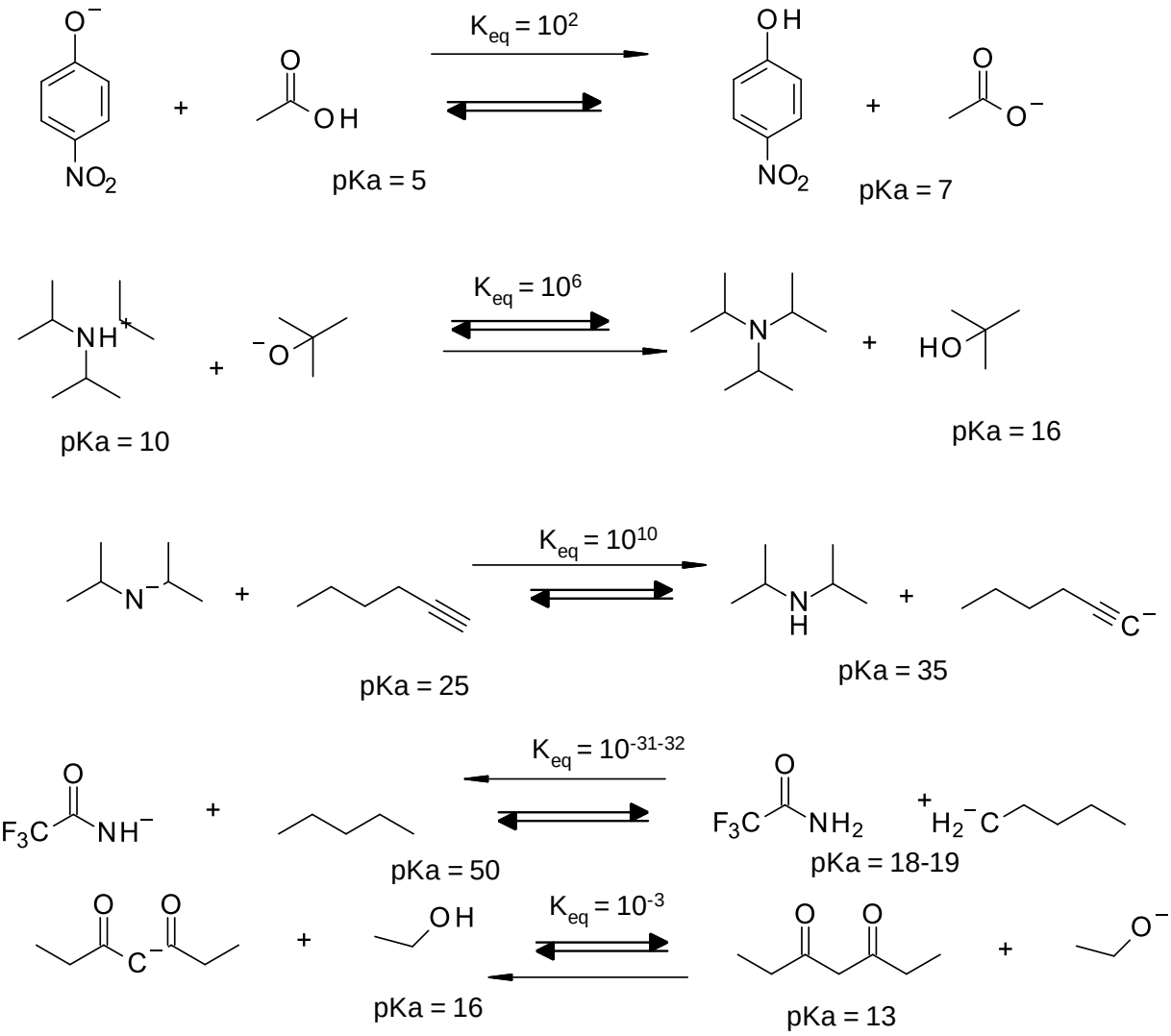
Problem 1 [10 marks]
Explain why compound B can be considered a doubly aromatic molecule.

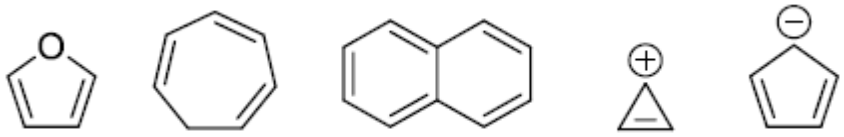


On the left hand side, the molecule is planar, cyclic, follow Hückel's rule (10 π electrons) and has every atom in the ring able to participate in delocalizing the electrons by having p-orbital or a lone pair of electrons. On the right hand side (after delocalization) the two cycles are aromatic (6 π electrons each). The molecule has two different aromaticity.

Problem 2 [10 marks]
Draw the products formed from the following acid base pairs and use approximate pKa values to predict in which direction the equilibrium will lie. (2 points for each reaction)



Problem 3 [10 marks]
Which of the following compounds are aromatic? Fully justify your answer. (5 compounds, 2 marks each).



In order to be aromatic a compound must

- Be cyclic
- Follow Hückel's rule (4n+2 π electrons)
- Be able to be planar
- Have every atom in the ring able to participate in delocalizing the electrons by having p-orbital or a lone pair of electrons

Starting from the left:

Aromatic 6 π electrons, planar, every atom involved

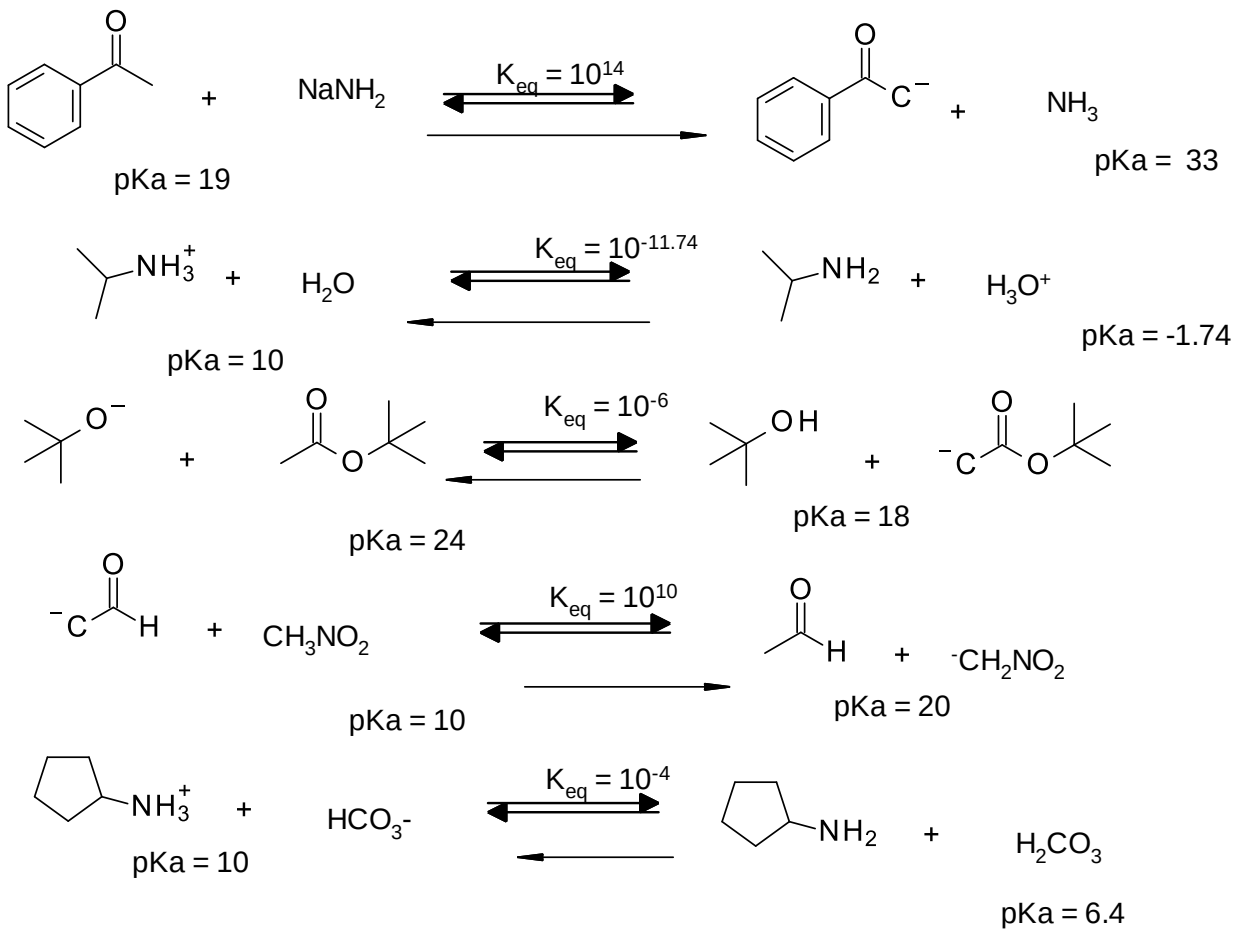
Not aromatic one sp3 carbon breaking the aromaticity

Aromatic 10 π electrons, planar, every atom involved

Aromatic 2 π electrons, planar, every atom involved

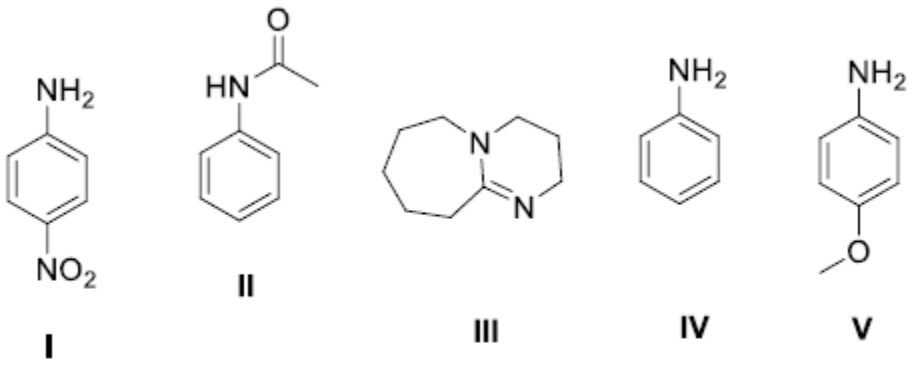
Aromatic 6 π electrons, planar, every atom involved

Problem 4 [10 marks]
Draw the products formed from the following acid-base pairs, using approximate pKa values to predict in which direction the equilibria will lie. (2 points each, 1 point for the product, one for the direction of the equilibrium)



Problem 5 [10 marks]

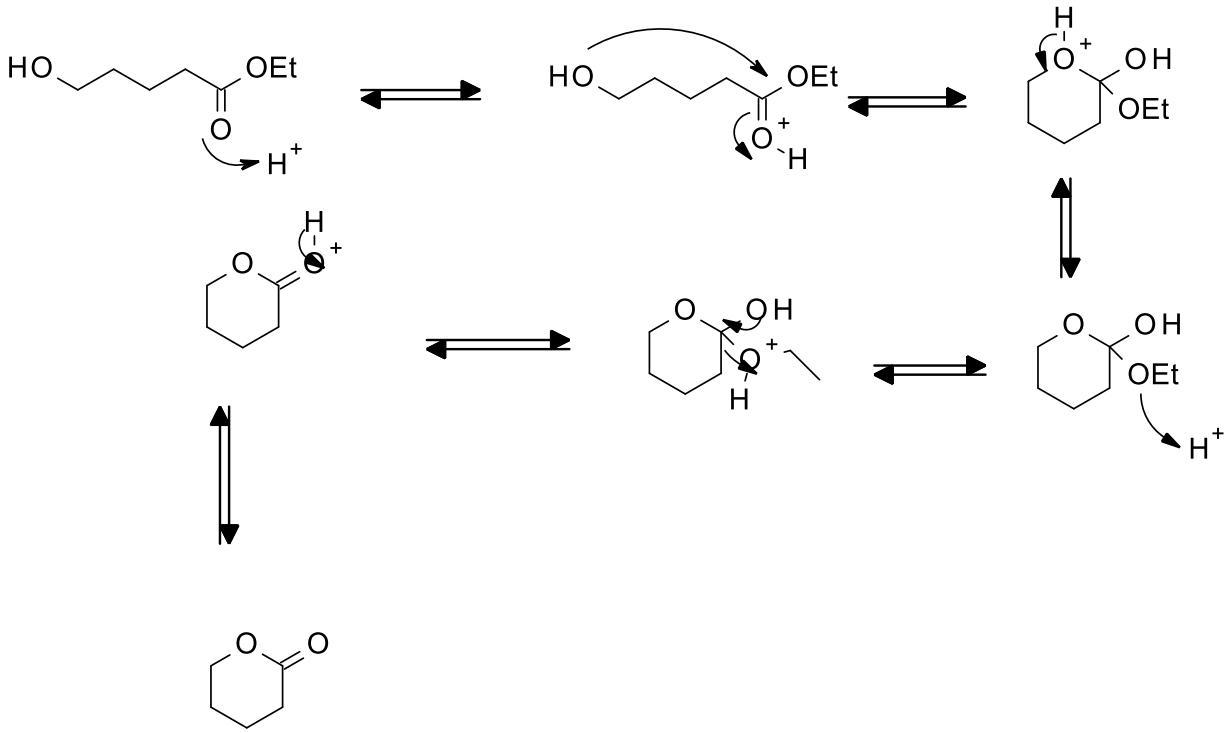
Rank compounds compounds I to V in order of increasing basicity. Explain your answers.

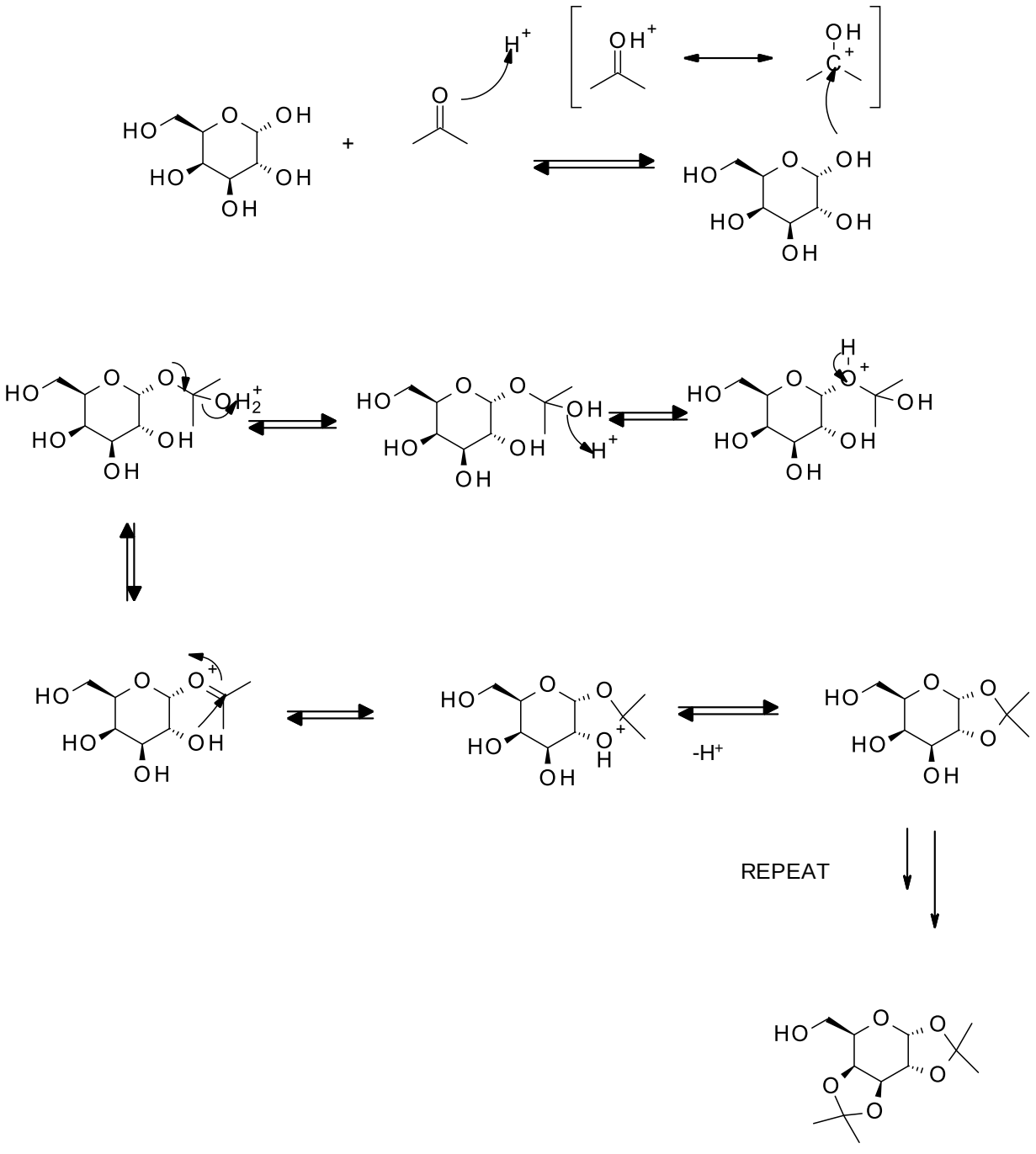


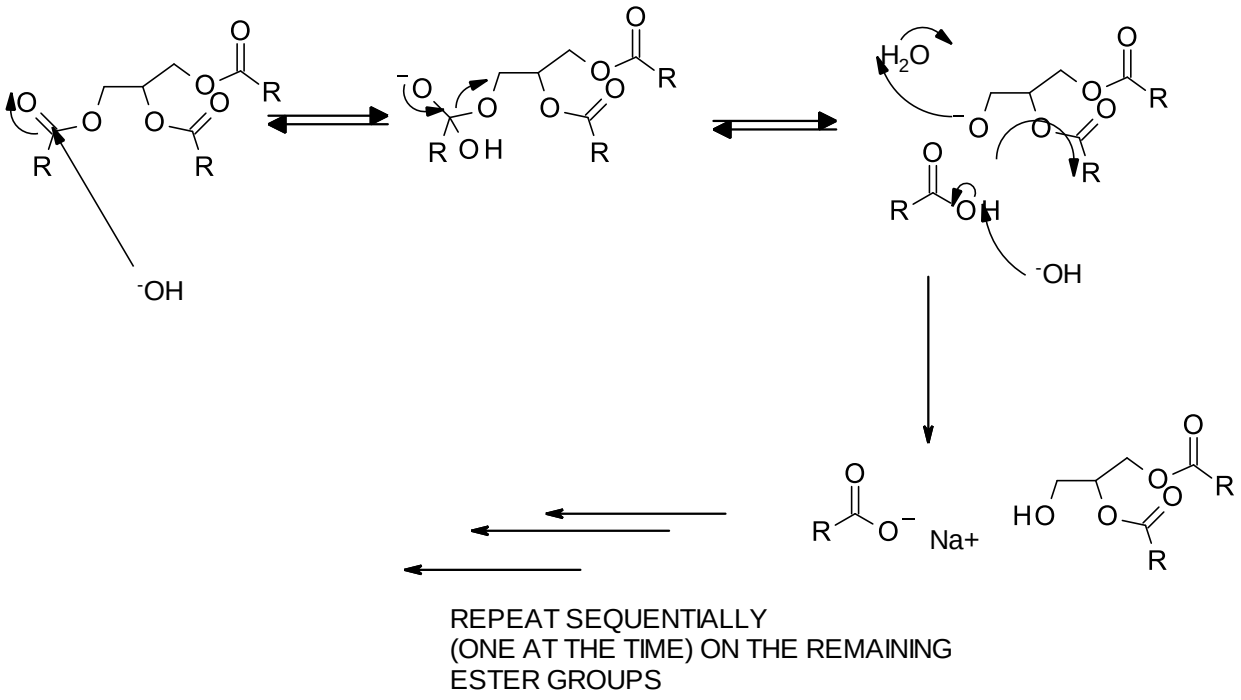
Amidine III is the most basic of all, acetamide II the least. I is less basic than IV while V is more basic than IV due to electron-withdrawing effect of NO_2 group and electron donating effect of OMe group for I and V respectively.

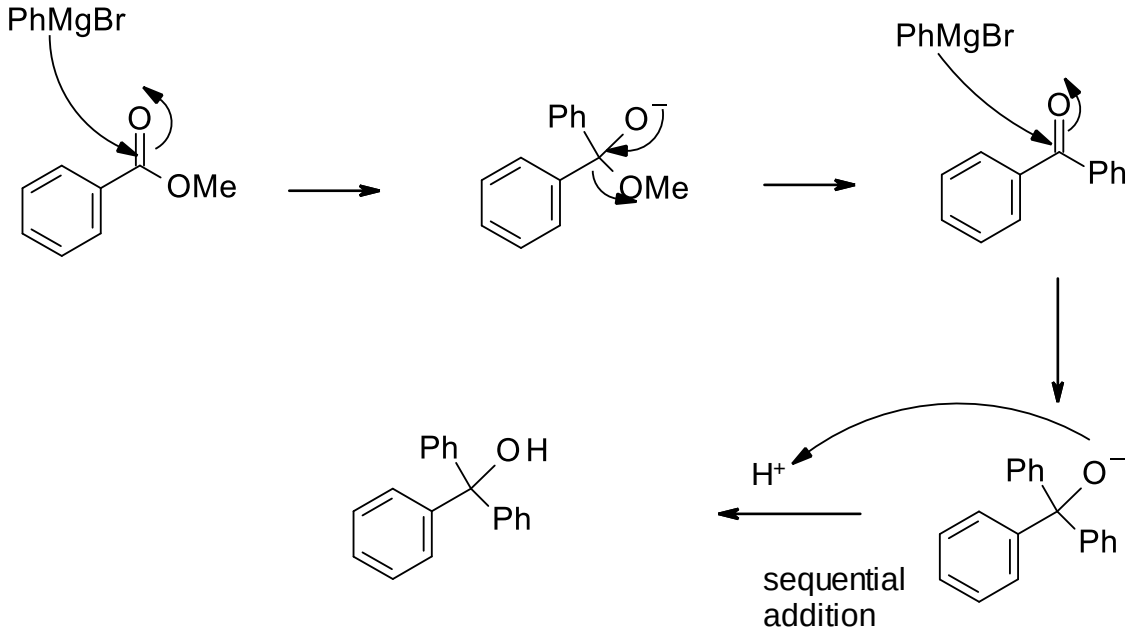
II < I < IV < V < III

Problem 6 [40 marks]
Provide a mechanism for the following reactions:



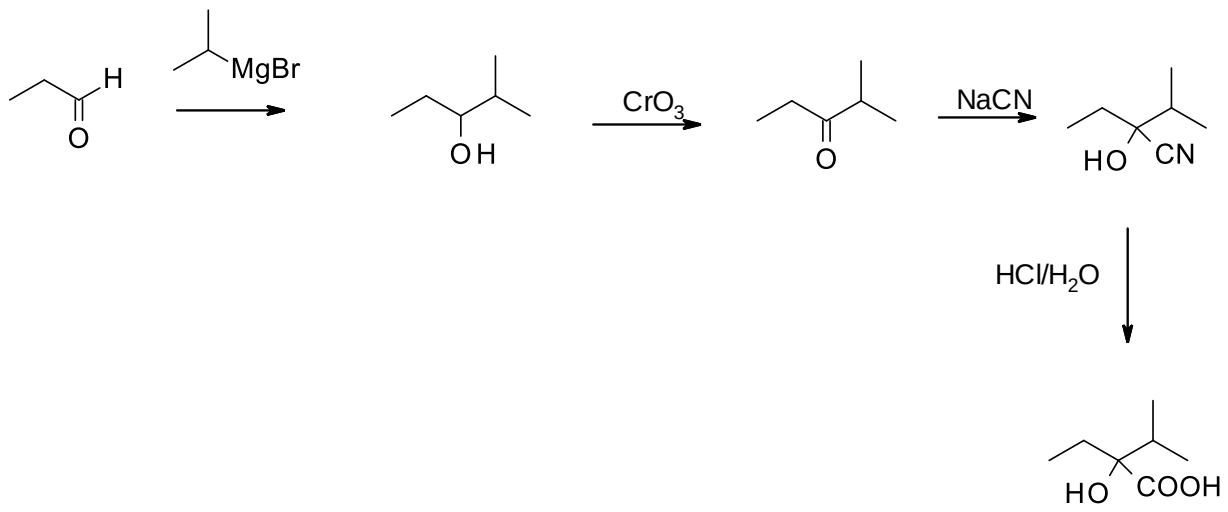

REPEAT





Problem 7 [10 marks]

Propose a multistep synthetic route leading to compound A by using nPrCHO (n-Butanal) and any other reagents you might need. No mechanisms required


H
O
MgBr
OH
CrO_3
O
NaCN
HO
CN
HCl/H_2O
HO
COOH